Name: Marie E. Rognes
Born: October 7 1982
Nationality: Norwegian
Present position: Chief Research Scientist, Simula Research Laboratory, Oslo, Norway

Education

2009	PhD in Applied Mathematics/Numerical Analysis (Date of conferred degree: 17.09.2009), Centre for Mathematics for Applications (Centre of Excellence), University of Oslo (UiO), Norway. Thesis title: *Mixed finite element methods for viscoelasticity and gels*
2005	MSc in Applied Mathematics, Department of Mathematics, UiO

Current and previous professional positions

2016–	Chief Research Scientist, Simula Research Laboratory, Oslo, Norway
2026–	Director, Centre for Digital Neurotechnology, Simula Research Laboratory, Oslo, Norway
2024–	Deputy Director, K. G. Jebsen Centre for Brain Fluid Research, University of Oslo, Norway
2022–2023	Fulbright Scholar, University of California San Diego, La Jolla, CA, USA
2020–2022	Professor II, Department of Mathematics, University of Bergen, Norway (20%)
2014–2017	Consultant, Melior Innovations, Houston, TX, USA (part-time)
2012–2016	Head of Biomedical Computing Department, Simula Research Laboratory
2015–2016	Associate Professor II, Department of Mathematics, UiO (20%)
2012–2016	Senior Research Scientist, Simula Research Laboratory, Norway
2012–2013	Lecturer, Department of Informatics, UiO (part-time)
2009–2012	Postdoctoral fellow, Center for Biomedical Computing, Simula Research Laboratory
Jan–Jun 2007	Visiting Scholar, University of Minnesota Twin Cities, Minneapolis, MN, USA
2005–2008	Research fellow, Centre for Mathematics for Applications, UiO

Awards, prizes, and academy memberships

- 2026 EMS/ECMI Lanczos Prize for Mathematical Software. This prize is awarded once annually by the European Mathematical Society and the European Consortium for Mathematics in Industry to recognize outstanding mathematical software. I won the 2026 prize together with my collaborators on the FEniCS core developer team, with the citation *for revolutionising the implementation of the finite element method*.
- Member of the Norwegian Academy of Technological Sciences (2022–).
- 2022 Fulbright Scholarship Award, U.S.-Norway Fulbright Foundation for Educational Exchange.
- 2018 Royal Norwegian Society of Sciences and Letters Prize for Young Researchers within the Natural Sciences. This prize is awarded once annually to researchers in Norway below the age of 40 who have accomplished exceptional results within their field, with documented originality and talent.
- 2015 Wilkinson Prize for Numerical Software. The Wilkinson prize for Numerical Software is the most prestigious prize in the numerical software research community, awarded once every four years to the authors of an outstanding piece of numerical software. I won the 2015 prize for the dolfin-adjoint project, together with my co-authors P. E. Farrell, S. W. Funke and D. A. Ham.
- Founding Member of the Young Academy of Norway (2015–2019). The Young Academy of Norway was initiated by the Norwegian Academy of Science and Letters and targets talented, young researchers interested in working interdisciplinarily with broader issues. Only 20 members were selected from all scientific disciplines out of 160 applicants in this first cycle of four-year terms.
- 2015 Simula Research Award. This award is presented annually to honour excellent contributions to research undertaken at Simula that lead to new insight within the field.

Research, innovation and mobility grants

2027–2031	FRIPRO Research Grant, Research Council of Norway (RCN): *BrainComplexes — From connectomes to continua: new complexes in brain multiphysics* (12 MNOK) (Project leader)
2026–2029	Emerging Technologies Research Grant, RCN: *DigiCells — Next-generation digital twins for biological cell environments* (12 MNOK) (Project leader)
2024–2026	Essential Open Source Software for Science Grant, Wellcome Trust: *Next-Generation Simulation and Learning in Imaging-Based Biomedicine* (0.4 MUSD) (Project leader)

2024–2029	K. G. Jebsen Centre for Medical Research, Foundation K. G. Jebsen: *K. G. Jebsen Centre for Brain Fluid Research* (61 MNOK) (Deputy Director)
2022–2026	UiO Life Science Convergence Environment: *AUTORHYTHM: an interdisciplinary research environment to elucidate the role of autophagy in healthy aging* (16 MNOK) (co-PI)
2021–2025	FRIPRO Research Grant (Researcher project for scientific renewal), RCN: *EMIx — Exciting times: Extreme modelling of excitable tissue* (12 MNOK) (Project leader)
2017–2023	ERC Starting Grant (PE1 Mathematics), European Research Council: *Mathematical and computational foundations for modelling cerebral fluid flow* (1.5 MEUR) (Project leader)
2016–2023	EPSRC CDT Oxford-Simula InFoMM Collaboration (∼0.1 MGBP) (co-PI)
2016–2019	FRIPRO Research Grant (Young Research Talent), RCN: *The Numerical Waterscape of the Brain* (8.7 MNOK) (Project leader)
2015–2018	Nordforsk Research Grant (the Nordic Council of Ministers): *Automated uncertainty quantification for numerical solution of partial differential equations* (6.7 MNOK) (Project leader)
2015–2016	Horizon 2020 MSCA RISE Research and Mobility Grant: *Unique Biomaterial-Drug Solution for Multifunctional Central Venous Catheters* (1.8 MNOK total) (Work package leader)
2014–2018	Horizon 2020 MSCA ITN Grant: *TheLink* (34 MNOK total) (Associated partner)
2012–2018	RCN Centre of Excellence Center for Biomedical Computing: Robust Solvers and High-level adjoints: software and applications (∼0.5 MEUR/year, Work package leader/PI)
2007	Leiv Eiriksson Programme Mobility Grant, RCN (60 KNOK) (Project leader)

The ∗ denotes upcoming events in the following.

Lectures, short courses and other teaching activities

2027∗	Invited lecture series at Advances in Computational Neurology: Mathematical Modeling, Simulation, and Data-Driven Methods, Lake Como School of Advanced Studies, Villa del Grumello, Italy, Sept 2027
2027∗	Invited lecture series, EMS School on Mathematical Modelling, Numerical Analysis and Scientific Computing, Kácov, the Czech Republic, June 6–11 2027
2024	Invited lecture series, 2024 Woudschoten Conference of the Dutch-Flemish Scientific Computing Society, the Netherlands, Sept 25–27 2024
2022	Finite elements and brain multiphysics, Invited Guest Lecture (80 min), MAE 207: Finite element analysis in coupled problems, UCSD, La Jolla, US
2021	Invited lecture series, Brain modelling: from magnetic resonance imaging to finite element simulation, Porous Media Math Short Course Series, University of Bergen (online)
2015, 2016	Lecturer, 3-day course on FEniCS and Dolfin-adjoint, EPSRC CDT NGCM Summer Academy, University of Southampton, UK (top-rated course of the Academy)
2015, 2022, 2026	Lecturer/Project mentor, Simula/UCSD Summer School in Computational Physiology
2015–2016	Lecturer, Solid mechanics (MEK2500, 10 ECTS), Department of Mathematics, UiO
2012–2013	Lecturer, Numerical methods (INF-MAT2351, 10 ECTS), Department of Informatics, UiO
2008–2013	Lecturer and demonstrator, FEniCS short courses, including at the American Institute of Mathematics, California, USA; Imperial College London, UK; King Abdullah University of Science and Technology, Saudi Arabia; and University of Southampton.
2001–2003	Assistant teacher in different courses in mathematics and computer science (introduction to programming, programming with high-level languages, numerical linear algebra) at the Faculty for Mathematics and Natural Sciences, University of Oslo.

Organization of scientific meetings (partial list)

2027*	12th International Conference on Computational Methods for Coupled Problems in Science and Engineering (COUPLED2027), Palma, Mallorca, Jun 2027 (Scientific committee)
2027*	Minisymposium, ICOSAHOM, Milan, Italy, 2027 (Co-organizer)
2027*	Invited Session, COUPLED2027, Palma, Spain, Jun 2027 (Organizer)
2027*	12th International Conference on Computational Methods for Coupled Problems in Science and Engineering, Palma, Mallorca, Jun 2027 (Scientific committee)
2019, 2027*	SIAM Conference on Computational Science and Engineering (SIAM CSE) (Scientific committee), both 2019 and 2027, by coincidence, in Spokane, US

2027*	ICIAM 2027: 11th International Congress on Industrial and Applied Mathematics, the Hague, the Netherlands, Jul 2027 (Program committee)
2026, 2027*	Platform for Advanced Scientific Computing (PASC) Conference, Co-chair/Chair for *Computational Methods and Applied mathematics*, Jun 16–18 2026, Basel, Switzerland
2026	Minisymposium, 17th World Congress on Computational Mechanics (WCCM)/10th ECCOMAS Congress, July 19-24, 2026, Münich, Germany (Co-organizer)
2020, 2025	Oslo Brain Fluid Symposium, Simula Research Laboratory, Norway (Chair)
2025	Workshop, *Multiphysics and mixed-dimensional modeling and discretizations*, Centre for Advanced Study, Norwegian Academy of Science and Letters, Nov 25–26 2025 (Co-chair)
2025	Minisymposium at ENUMATH 2025, Sep 1–5 2025, Heidelberg, Germany (Co-organizer)
2025	Minisymposium at COUPLED2025, May 25–28 2025, Sardinia, Italy (Co-organizer)
2016, 2024	FEniCS Conference, Simula Research Laboratory, Oslo, Norway (Organizer)
2024	13th European Conference on Mathematical and Theoretical Biology, July 2024, Toledo, Spain (Scientific Committee)
2023	Minisymposium at SIAM CSE'23, Amsterdam, the Netherlands (Co-organizer)
2022	ECCOMAS Congress 2022, 8th European Congress on Computational Methods in Applied Sciences and Engineering, Norway (Scientific committee/Young Investigators Initiative Chair)
2022	Minisymposium at ECCOMAS 2022, *Advances in automatic code-generation software for simulations in science and engineering*, Oslo, Norway (Co-organizer)
2019	Minisymposium at ENUMATH 2019, *Discretizations of mixed-dimensional PDEs*, Egmond aan Zee, the Netherlands (Organizer)
2019	5th Cerebrospinal Fluid Dynamics Symposium, Simula, Norway (Co-chair)
2018	PDESOFT 2018, Bergen, Norway (Scientific committee)
2018	Minisymposium at ECCM-ECFD 2018, *Mathematical and computational modelling of fluid flow and transport in the brain and spinal cord*, Glasgow, UK (Organizer)
2015	Minisymposium at SIAM CSE'15, *Efficient Solvers for PDE-Constrained Optimization minisymposium*, Salt Lake City, UT, USA (Co-organizer)
2013	26th Nordic Seminar on Computational Mechanics, Simula Research Laboratory, Norway

Major scientific software projects

2007–	Core FEniCS developer
2011–	Founding member and core developer of the Dolfin-adjoint project
2016–2024	Founding member of the FEniCS Steering Council

Editorial board memberships

2023–	Editorial board member, European Mathematical Society Magazine
2022–	Associate editor, AIMS: Applied Mathematics for Modern Challenges (AIMS)
2022–	Associate editor, ESAIM: Mathematical Modelling and Numerical Analysis (ESAIM: M2AN)
2020–	Editorial board member, Springer Nature Partial Differential Equations and Applications
2020–	Editorial board member, BMC Fluids and Barriers of the Central Nervous System
2020–2026	Associate editor, SIAM Journal of Scientific Computing (SISC)
2020–2023	Topic editor, Journal of Open Source Software (JOSS)

Invited research programmes and other longer-term mobility

2027*	*Polytopal Finite Element Methods for PDEs: Theory, Algorithms, and Machine Learning Enhancement*, Mathematisches Forschungsinstitut Oberwolfach, Oct 3-8 2027
2025	*Interfaces and unfitted discretization methods*, Mittag-Leffler Institute, Stockholm, Sweden (1 month, Sep 2027)
2022–2023	Visiting Fulbright Scholar at the Institute for Engineering in Medicine, University of California San Diego, San Diego, USA (11 months, Aug–Jul)
2012	Visiting Scholar at the *Multiscale Numerics for the Atmosphere and Ocean* research programme, Isaac Newton Institute, Cambridge, UK (1 month, Dec 2012).
2007	Visiting Scholar at the Institute for Mathematics and its Applications, University of Minnesota Twin Cities, Minneapolis, USA (6 months, Jan–Jun).

Other invited and elected memberships and positions

2024–2028	Research Council of Norway Portfolio Board for Ground-breaking Research
2023–2026	European Mathematical Society Committee for Applications and Interdisciplinary Relations
2023–2027	Interpore Council

Society memberships

2022–	European Mathematical Society (EMS)
2022–2023	Society for Neuroscience (SfN)
2014, 2022–2023, 2026–	Society for Industrial and Applied Mathematics (SIAM)

Reviewer for international journals

I have acted as a reviewer for the following scientific journals

SIAM Journal on Applied Mathematics, SIAM Journal on Scientific Computing, Mathematics of Computation, Mathematical Models and Methods in Applied Sciences, BIT Numerical Mathematics, ACM Transactions on Mathematical Software, Journal of Numerical Mathematics, Annals of Biomedical Engineering, International Journal on Numerical Methods in Biomedical Engineering, Nature Scientific Reports, PLOS Computational Biology, Fluids and Barriers of the Central Nervous System, Journal of Fluid Mechanics, Journal of Computational Physics, and Geoscientific Model Development.

Scientific advisory boards

- ◇ Member of the Scientific advisory board for the Research Council of Finland Flagship Programme's Flagship of Advanced Mathematics for Sensing, Imaging and Modelling (FAME) (https://fameflagship.fi/).
- ◇ Member of the International Advisory Board, Center for Computational and Stochastic Mathematics (CEMAT), Instituto Superior Técnico, Lisbon, Portugal (https://cemat.tecnico.ulisboa.pt/main.php).

Evaluation committees

- ◇ Member of **PhD thesis evaluation committees** at INRIA/Université Claude Bernard Lyon 1, France (2026∗); Norwegian University of Life Science (2026); University of Zürich, Switzerland (2025), LSM Ecole Polytechnique, Paris, France (2023); the MOX, Politechnico di Milano, Italy (2018, 2020, 2024, 2024); Department of Mathematics and Computer Science, University of Southern Denmark, Odense (2018); Department of Computer Science, Imperial College London (2014); Department of Informatics, University of Grenoble, France (2013); and the Department of Informatics, University of Bergen, Norway (2012);
- ◇ Member of **faculty evaluation committees** at Department of Information Technology, University of Uppsala (2025); Department of Mathematics, University of Uppsala (2020); Department of Mathematics, Oslo and Akershus University College (2014); Departments of Informatics, Oslo and Akershus University College (2014); Department of Informatics, University of Oslo (2001).
- ◇ European Research Council, **panel member** (ERC–STG–PE1: 2024, 2026) and **external reviewer** (2016–).
- ◇ Swedish Research Council, **panel member** (2024, 2025, 2026).

Supervision of postdoctoral fellows and graduate students

Postdoctoral fellows

PD8) Dr. Marte Julie Sætra studied *modelling and simulation of electrodiffusion and interstitial fluid flow* as a Postdoctoral Fellow with me at Simula (2021–2024); Sætra is now a Senior Scientist at the Norwegian Defence Research Establishment.

PD7) Dr. Rami Masri studied *mixed-dimensional modelling of excitable tissue* as a Postdoctoral Fellow with me at Simula (2022–2024). Masri then accepted a Postdoctoral Research Fellowship at Brown University, Providence, RI, USA; and is now an Assistant Professor (tenure-track) at Penn State University, PA, USA.

PD6) Dr. Pietro Benedusi studied *extreme scale modelling and simulation of excitable tissue* as a Postdoctoral Fellow with me at Simula (2022–2024). Benedusi is now a Lecturer at the Università della Svizzera Italiana, Lugano, Switzerland;

PD5) Dr. Ingeborg Gjerde studied *predictive modelling of perivascular flows* as a Postdoctoral Fellow in the Waterscales (ERC) project at Simula (2020–2022), was subsequently employed as a Research Engineer at Simula, and is now in R&D at the Norwegian Geotechnical Institute;

PD4) Dr. Cécile Daversin-Catty studied *mixed dimensional mathematical and computational modelling* as a Postdoctoral Fellow in the Waterscales (ERC) project (2017–2021), and was subsequently employed as a Research Scientist at Simula;

PD3) Dr. Travis B. Thompson studied *mixed finite element methods for multiple-network poroelasticity* as a Postdoctoral Fellow in the Waterscape (RCN) project (2017–2019). Thompson was next a Postdoctoral Research Associate at the Department of Mathematics, University of Oxford, before accepting a faculty position at Texas Tech University, Texas, US.

PD2) Dr. Rocio Rodriguez-Cantano studied *uncertainty quantification in cardiac electromechanics* as a Postdoctoral Fellow in the AUQ-PDE (NordForsk) project (2016–2018);

PD1) Dr. Simon Funke studied *predictive modeling in physiological flow* as a postdoctoral fellow with me at Simula (2013–2015). Funke was subsequently awarded a FRIPRO Young Research Talent Grant, became a Senior Research Scientist and Research Director at Simula, before launching a deep technology start-up company.

PhD students

P19) Mr. Jakob L. Sønstebø is currently pursuing his PhD in computational mathematics on the topic of *innovative algorithms for simulation of excitable tissue* under my supervision at Simula.

P18) Mr. Sami Laubo is currently pursuing his PhD in computational mathematics on the topic of *Deep learning for modeling brain fluids* under my co-supervision (with K.-A. Mardal as main supervisor) at Simula.

P17) Ms. Lucie Wintrová is currently pursuing her PhD in at the Mathematical Institute, Charles University, Prague, the Czech Republic on the topic of *Homogenization in models of neurodegeneration* under my co-supervision (with M. Bulíček as main supervisor).

P16) Ms. Eirill Strand Hauge is currently pursuing her PhD on the topic of *cellular modelling of electrochemical interactions in neural tissue* under my supervision at Simula.

P15) Ms. Ingvild Devold is currently pursuing her PhD on the topic of *Mechanochemical interplay underlying human performance* under my supervision (with P. Rangamani, UCSD as co-supervisor) at Simula.

P14) Ms. Nanna Berre is currently pursuing her PhD on the topic of *cut finite element methods for modelling excitable tissue* under my co-supervision (with A. Massing as main supervisor) at the Norwegian University of Science and Technology.

P13) Mr. Balint Laczko successfully completed his PhD in Music technology on the topic of *Instrumental image sonification* under my co-supervision (with A. Jensenius as main supervisor) at RITMO, University of Oslo in August 2026.

P12) Mr. Marius Causemann successfully completed his PhD on the topic of *Multi-Scale Modelling of Intracranial Biomechanics and Solute Transport* under my supervision at Simula and the UiO in January 2026.

P11) Dr. Martin Hornkjøl successfully completed his PhD on the topic of *Modelling fluid flow and solute transport in the intracranial compartment and developing methods for multi-physics problems* under my co-supervision (with K. A. Mardal as main supervisor) at the University of Oslo in 2025.

P10) Dr. Georgia Brennan successfully completed her DPhil on the topic of *Mathematical modelling of clearance and proteopathy in neurodegenerative diseases with application to Alzheimer's disease* under my co-supervision (with A. Goriely as main supervisor) at the University of Oxford in March 2024.

P9) Dr. Eleonora Piersanti completed her PhD on *Parameter-robust formulation and preconditioning of poroelasticity equations for brain modelling* under my supervision at Simula in 2023. Piersanti is now a software and data science consultant at Expert Analytics, Oslo, Norway.

P8) Dr. Nicolas Boullé completed his DPhil under my co-supervision (with P. E. Farrell as main supervisor) at the University of Oxford in 2022. He was later a Simons Postdoctoral Research Fellow in Mathematics at the University of Cambridge, and is now an Assistant Professor in Applied Mathematics at Imperial College London.

P7) Dr. Ada J. Ellingsrud completed her PhD on *Computational modelling of electrodiffusion and osmosis in cerebral tissue* under my supervision at Simula in 2021. Ellingsrud is now a Research Scientist at Simula, Coordinator for the Simula@BI Research Centre, and Head of Simula's Scientific Computing and Numerical Analysis Department.

P6) Dr. Lars M. Valnes completed his PhD on *Patient-specific Modeling of the Human Brain using Magnetic Resonance Imaging* under my co-supervision (with K.-A. Mardal as main supervisor) at the University of

Oslo in 2020. Valnes was next a Research Engineer at Oslo University Hospital – Rikshospitalet, Norway.

P5) Dr. Vegard Vinje completed his PhD on *Mathematical Modeling of Cerebrospinal Fluid Pulsatility and Pathways* under my supervision at Simula in 2019. Vinje was next a Research Scientist and Research Director for Scientific Computing at Simula, before taking on the role of Technical Lead for Models and Algorithms at the major renewable energy company Hafslund.

P4)) Dr. Matteo Croci completed his DPhil (PhD) in Mathematics on *Multilevel Monte Carlo Methods for Uncertainty Quantification in Brain Simulations* under my co-supervision (with P. E. Farrell and M. Giles as main supervisors) at the University of Oxford in 2019. Croci was next a Postdoctoral Research Fellow at the Oden Institute, University of Texas at Austin, US, and is now an Ikerbasque Research Fellow (tenure-track) and Ramón y Cajal Fellow at the Basque Center for Applied Mathematics (BCAM).

P3) Dr. Gabriel Balaban completed his PhD on *Adjoint Data Assimilation Methods for Cardiac Mechanics* under my supervision at Simula in 2017. He subsequently won a Postdoctoral Research Associate position at King's College London, UK and a Postdoctoral Fellowship in Bioinformatics at the University of Oslo; he is now Associate Professor at Kristiania University of Applied Sciences, Oslo, Norway.

P2) Dr. Giulia Pizzichelli completed part of her PhD studies with me (and with E. Sinibaldi as main supervisor) at Simula in 2015 and graduated from the School of Advanced Studies, Pisa/Istituto Italiano di Tecnologia, Italy in 2016.

P1) Dr. Andre Massing completed his PhD on *Analysis and Implementation of Finite Element Methods on Overlapping and Fictitious Domains* under my co-supervision (with A. Logg as main supervisor) at Simula in 2012. He was subsequently offered an Assistant Professorship at Umeå University, Sweden, and is now an Associate Professor at the Norwegian University of Technology and Science, Trondheim, Norway.

MSc students

I have supervised or actively co-supervised the following MSc students in their thesis research at the Departments of Informatics and/or Mathematics at the UiO, at the Norwegian University of Technology and Science, Trondheim, Norway, at the University of Trento, Italy, and at Karlsruhe Institute of Technology, Germany:

Albert Bayazidi (2026), Sami Laubo (2025), Åsmund Aamodt Resell (2022), Nanna Berre (2022), Marius Causemann (2020), Emilie Ødegaard (2018), Valentyna Pysarieva (2018), Carlo Cisale (2017), Janis Grobovs (2017), Vegard Vinje (2016), Ada J. Ellingsrud (2015), Ingeborg Sauge Torpe (2014), Nina Ødegaard (2014), Eline Sundt (2012), Fredrik Valdmanis (2012),

Other official mentorships

◇ Mentor for the COMET – Coaching, Mentoring and Training at the University of Bern career development programme (Sep 2025-Feb 2026)

Preprints

68. A. Solheim, G. Ringstad, P. K. Eide, M. E. Rognes, E. Acar, and K.-A. Mardal, Low-rank tensor decompositions reveal coupled spatiotemporal patterns of CSF tracer transport in the human brain, in review, 2026
67. B. Laszkó, H. Knævelsrud, M. E. Rognes, and A. Refsum Jensenius, Image sonification of biological data: a systematic review, in review, 2026
66. G. S. Brennan, T. B. Thompson, H. Oliveri, V. Vinje, G. Ringstad, P. K. Eide, A. Goriely and M. E. Rognes, One night of sleep deprivation reduces spatially patterned human brain clearance: implications for simulated tau propagation, in review, 2026

Articles in peer-reviewed international journals and select preprints

65. I. S. Devold, M. E. Rognes, P. Rangamani, Mechanochemical modeling of exercise-induced skeletal muscle hypertrophy, PLoS Computational Biology, 2026
64. M. Causemann, M. Kuchta, R. Masri, and M. E. Rognes, In-silico solute transport via perivascular networks in the human intracranial space, Nature Communications, 2026
63. M. Causemann, R. Enger and M. E. Rognes, Stretch and flow at the gliovascular interface: high-fidelity modelling of astrocyte endfeet, Proceedings of the National Academy of Sciences, 2026
62. I. A. Baratta, J. P. Dean, J. S. Dokken, M. Habera, J. S. Hale, C. N. Richardson, M. E. Rognes, M. W. Scroggs, N. Sime and G. N. Wells, DOLFINx: the next generation FEniCS problem solving environment, Zenodo, 2025

61. H. Herlyng, A. J. Ellingsrud, M. Kuchta, I. Jeong, M. E. Rognes and N. Jurisch-Yaksi, Advection versus diffusion in brain ventricular transport, Fluids and Barriers of the CNS, 2025
60. E. A. Francis, J. G. Laughlin, J. S. Dokken, H. N. T. Finsberg, C. T. Lee, M. E. Rognes and P. Rangamani, Spatial modeling algorithms for reactions and transport in biological cells, Nature Computational Science, 2024
59. L. W. Dreyer, A. Eklund, M. E. Rognes, J. Malm, S. Qvarlander, K.-H. Støverud, K.-A. Mardal and V. Vinje, Modeling CSF circulation and the glymphatic system during infusion using subject specific intracranial pressures and brain geometries, Fluids and Barriers of the Central Nervous System, 2024
58. I. Gjerde, M. Kuchta, M. E. Rognes and B. Wohlmuth, Directional flow in perivascular networks: mixed finite elements for reduced-dimensional models on graphs, Journal of Theoretical Biology, 2024
57. P. Benedusi, A. J. Ellingsrud, H. Herlyng, M. E. Rognes, Scalable approximation and solvers for ionic electrodiffusion in cellular geometries, SIAM Journal on Scientific Computing, 2024
56. R. Masri, M. Zeinhofer, M. Kuchta and M. E. Rognes, The modelling error in multi-dimensional time-dependent solute transport models, ESAIM: M2AN, 2024.
55. N. Berre, M. E. Rognes and A. Massing, Cut finite element discretizations of cell-by-cell EMI electrophysiology models, SIAM Journal on Scientific Computing, 2024
54. P. Benedusi, P. Ferrari, M. E. Rognes and S. Serra-Capizzano, Modeling excitable cells with the EMI equations: spectral analysis and iterative solution strategy, Journal of Scientific Computing, 2024
53. I. Gjerde, M. E. Rognes and A. Sánchez, The directional flow generated by peristalsis in perivascular networks – theoretical and numerical reduced-order description, Journal of Applied Physics, 2023
52. E. Piersanti, M. E. Rognes, V. Vinje, Are brain displacements and pressures within the parenchyma induced by surface pressure differences? A computational modelling study, PLOS ONE, 2023.
51. J. G. Laughlin, J. S. Dokken, H. N. Finsberg, E. A. Francis, C. T. Lee, M. E. Rognes and P. Rangamani, SMART: Spatial Modeling Algorithms for Reaction and Transport, Journal of Open Source Software, 2023
50. V. Vinje, B. Zapf, G. Ringstad, P.-K. Eide, M. E. Rognes, K.-A. Mardal, Human brain solute transport quantified by glymphatic MRI-informed biophysics during sleep and sleep deprivation, Fluids and Barriers of the Central Nervous System, 2023.
49. M. J. Sætra, A. J. Ellingsrud and M. E. Rognes, Neural activity induces strongly coupled electro-chemo-mechanical interactions and fluid flow in astrocyte networks and extracellular space – a computational study, PLOS Comp. Bio., 2023.
48. E. Eliseussen, M. E. Rognes, T. B. Thompson, A-posteriori error estimation and adaptivity for multiple-network poroelasticity, ESAIM: M2AN, 2023.
47. G. S. Brennan, T. B. Thompson, H. Oliveri, M. E. Rognes, A. Goriely, The role of clearance in neurodegenerative disease, SIAM Journal on Applied Mathematics, 2023.
46. N. Boullé, P. E. Farrell, M. E. Rognes, Optimal control of Hopf bifurcation points, SIAM Journal on Scientific Computing, 2023.
45. M. E. Rognes, Waterscales: Mathematical and computational foundations for cerebral fluid flow. European Mathematical Society Magazine, 2022
44. Q. Hong, J. O. Kraus, M. I. Kuchta, M. A. Lymbery, K.-A. Mardal, M. E. Rognes, Robust approximation of generalized Biot-Brinkman problems. Journal of Scientific Computing, Springer, 2022.
43. M. Causemann, V. Vinje and M. E. Rognes, Human intracranial pulsatility during the cardiac cycle: a computational modelling framework, Fluids and Barriers of the Central Nervous System, 2022
42. M. Hornkjøl, L. M. Valnes, G. Ringstad, M. E. Rognes, P. K. Eide, K.-A. Mardal and V. Vinje, CSF circulation and dispersion yield rapid clearance from intracranial compartments, Frontiers in Bioengineering and Biotechnology, 2022.
41. C. Daversin-Catty, I. G. Gjerde, and M. E. Rognes, Geometrically reduced modelling of pulsatile flow in perivascular networks, Frontiers in Physics, 2022.
40. A. J. Ellingsrud, D. B. Dukefoss, R. Enger, G. Halnes, K. Pettersen and M. E. Rognes, Validating a computational framework for ionic electrodiffusion with cortical spreading depression as a case study, eNeuro, 2022.
39. X. Lai, H. A. Taskén, T. Mo, S. W. Funke, A. Frigessi, M. E. Rognes and A. Köhn-Luque. A scalable solver for a stochastic, hybrid cellular automaton model of personalized breast cancer therapy. International

Journal on Numerical Methods for Biomedical Engineering e3542, 2021

38. A. J. Ellingsrud, N. Boullé, P. E. Farrell and M. E. Rognes. Accurate numerical simulation of electrodiffusion and water movement in brain tissue. IMA Mathematical Medicine & Biology, 2021
37. C. Daversin-Catty, C. N. Richardson, A. J. Ellingsrud and M. E. Rognes. Abstractions and automated algorithms for mixed-dimensional finite element methods. ACM Transactions on Mathematical Software, 2021
36. V. Vinje, E. N. Bakker and M. E. Rognes. Brain solute transport is more rapid in periarterial than perivenous spaces. Nature Scientific Reports, 2021
35. E. Piersanti, J. J. Lee, K.-A. Mardal, M. E. Rognes and T. Thompson. Parameter robust preconditioning by congruence for multiple-network poroelasticity. SIAM Journal on Scientific Computing, 2021
34. K.-A. Mardal, M. E. Rognes and T. B. Thompson. Accurate discretization of poroelasticity without Darcy stability – Stokes-Biot revisited. BIT Numerical Mathematics, 2021.
33. C. Daversin-Catty, V. Vinje, K.-A. Mardal and M. E. Rognes. The mechanisms behind perivascular fluid flow. PLOS ONE, 2020.
32. V. Vinje, A. Eklund, K.-A. Mardal, M. E. Rognes and K.-H. Støverud. Intracranial pressure elevation alters CSF clearance pathways. Fluids and Barriers of the CNS, 2020.
31. M. Croci, V. Vinje and M. E. Rognes. Fast uncertainty quantification of tracer distribution in the brain interstitial fluid with multilevel and quasi Monte Carlo. International Journal for Numerical Methods in Biomedical Engineering, 2020.
30. A. J. Ellingsrud, A. Solbraa, G. T. Einevoll, G. Halnes and M. E. Rognes. Finite element simulation of ionic electrodiffusion in cellular geometries. Frontiers in Neuroinformatics, 2020
29. M. Croci, V. Vinje and M. E. Rognes. Uncertainty quantification of parenchymal tracer distribution using random diffusion and convective velocity fields. Fluids and Barriers of the CNS, 2019
28. X. Lai, O. Geier, T. Fleischer, Ø. Garred, E. F. Borgen, S. Funke, S. Kumar, M. E. Rognes, T. Seierstad, A.-L. Børresen-Dale, V. N. Kristensen, O. Engebraaten, A. Kohn-Luque and A. Frigessi. Towards personalized computer simulation of breast cancer treatment: a multi-scale pharmacokinetic and pharmacodynamic model informed by multi-type patient data. Cancer Research, 2019
27. V. Vinje, G. Ringstad, E. K. Lindstrom, L. M. Valnes, M. E. Rognes, P. K. Eide and K.-A. Mardal. Respiratory influence on cerebrospinal fluid flow – a computational study based on long-term intracranial pressure measurements. Nature Scientific Reports, 2019
26. P. E. Farrell, J. E. Hake, S. W. Funke and M. E. Rognes. Automated adjoints of coupled PDE-ODE systems. SIAM Journal on Scientific Computing, 2019.
25. J. J. Lee, E. Piersanti, K.-A. Mardal and M. E. Rognes. A mixed finite element method for nearly incompressible multiple-network poroelasticity. SIAM Journal on Scientific Computing, 2019.
24. R. Rodriguez-Cantano, J. Sundnes and M. E. Rognes. Uncertainty in cardiac myofiber orientation and stiffnesses dominate the variability of left ventricle deformation response. International Journal for Numerical Methods in Biomedical Engineering, 2019.
23. M. Croci, M. B. Giles, M. E. Rognes and P. E. Farrell. Efficient white noise sampling and coupling for multilevel Monte Carlo with non-nested meshes. SIAM Journal on Uncertainty Quantification, 2018
22. V. Vinje, J. Bruckner, M. E. Rognes, K.-A. Mardal and V. Haughton. Fluid dynamics in syringomyelia cavities: Effects of heart rate, CSF velocity, CSF velocity waveform and craniovertebral decompression. The Neuroradiology Journal, 2018
21. G. Balaban, H. Finsberg, S. Funke, T. F. Håland, E. Hopp, J. Sundnes, S. Wall and M. E. Rognes. Data assimillation allows for in-silico identification of cardiac elastic heterogeneity in an infarcted human. Biomechanics and Modeling in Mechanobiology, 2018
20. G. Pizzichelli, B. Kehlet, Ø. Evju, B. Martin, M. E. Rognes, K.-A. Mardal and E. Sinibaldi. Numerical study of intrathecal drug delivery to a permeable spinal cord: effect of catheter position and angle. Computer Methods in Biomechanics and Biomedical Engineering, 2017
19. A. Tveito, K. H. Jæger, M. Kuchta, K.-A. Mardal and M. E. Rognes. A cell-based framework for numerical modelling of electrical conduction in cardiac tissue. Frontiers in Physics, Computational Physics, 2017
18. S. Kallhovd, M. M. Maleckar and M. E. Rognes. Inverse estimation of cardiac activation times via gradient-based optimisation. International Journal for Numerical Methods in Biomed. Engineering, 2017
17. M. E. Rognes, P. E. Farrell, S. W. Funke, J. E. Hake and M. M. C. Maleckar. cbcbeat: an adjoint-enabled

framework for computational cardiac electro-physiology. Journal of Open Source Software, 2017

16. G. Balaban, H. Finsberg, H. H. Odland, M. E. Rognes, S. Ross, J. Sundnes and S. Wall. High resolution data assimilation of cardiac mechanics applied to a dyssynchronous ventricle. International Journal for Numerical Methods in Biomedical Engineering, 2017
15. G. Balaban, M. S. Alnæs, J. Sundnes and M. E. Rognes. Adjoint multi-start based estimation of cardiac hyperelastic material parameters using shear data. Biomechanics and Modeling in Mechanobiology, 2016
14. M. Alnæs, J. Blechta, J. Hake, A. Johansson, B. Kehlet, A. Logg, C. Richardson, J. Ring, M. E. Rognes and G. N. Wells. The FEniCS Project Version 1.5. Archive of Numerical Software, 2015
13. A. Massing, M. G. Larson, A. Logg and M. E. Rognes. A Nitsche-based cut finite element method for a fluid-structure interaction problem. Communications in Applied Mathematics and Comput. Science, 2015
12. A. Massing, M. G. Larson, A. Logg and M. E. Rognes. A stabilized Nitsche overlapping mesh method for the Stokes problem. Numerische Mathematik, 2014
11. A. Massing, M. G. Larson, A. Logg and M. E. Rognes. A stabilized Nitsche fictitious domain method for the Stokes problem. Journal of Scientific Computing, 2014
10. M. S. Alnæs, A. Logg, K. B. Ølgaard, M. E. Rognes and G. N. Wells. Unified Form Language: A domain-specific language for weak formulations of partial differential equations. ACM Transactions on Mathematical Software, 2014
9. M. E. Rognes, D. A. Ham, C. J. Cotter and A. T. T. McRae. Automating the solution of PDEs on the sphere and other manifolds in FEniCS 1.2. Geoscientific Model Development, 2013
8. P. E. Farrell, D. A. Ham, S. W. Funke and M. E. Rognes. Automated derivation of the adjoint of high-level transient finite element programs. SIAM Journal on Scientific Computing, 2013.
7. M. E. Rognes and A. Logg. Automated goal-oriented error control I: stationary variational problems. SIAM Journal on Scientific Computing, 2013
6. L. Vynnytska, M. E. Rognes and S. R. Clark. Benchmarking FEniCS for mantle convection simulations. Computers & Geosciences, 2013.
5. A. Tveito, G. T. Lines, M. E. Rognes and M. M. Maleckar. An analysis of the shock strength needed to achieve defibrillation in a simplified mathematical model of cardiac tissue. International Journal of Numerical Analysis and Modeling, 2012.
4. M. E. Rognes and R. Winther. Mixed finite element methods for linear viscoelasticity with weak symmetry. Mathematical Models and Methods in Applied Science, 2010.
3. M. E. Rognes, M. C. Calderer and C. A. Micek. Modelling of and mixed finite element methods for gels in biomedical applications. SIAM Journal of Applied Mathematics, 2009.
2. M. E. Rognes, R. C. Kirby and A. Logg. Efficient assembly of H(div) and H(curl) conforming finite elements. SIAM Journal on Scientific Computing, 2009.
1. D. N. Arnold and M. E. Rognes. Stability of Lagrange elements for the mixed Laplacian. Calcolo, 2009.

Peer-reviewed books and edited volumes

4. J. S. Dokken, H. N. Finsberg, J. Hale, M. E. Rognes, and M. W. Scroggs. The FEniCS Project: The FEniCS 2024 Conference, Simula SpringerBriefs in Computing, Springer Nature, 2026.
3. J. S. Dokken, K.-A. Mardal, M. E. Rognes, L. M. Valnes and V. Vinje. Mathematical modeling of the human brain II: from glymphatics to deep learning, Simula SpringerBriefs in Computing, Springer Nature, 2026.
2. K.-A. Mardal, M. E. Rognes, T. B. Thompson and L. M. Valnes. Mathematical modeling of the human brain: from magnetic resonance images to finite element simulation, Simula SpringerBriefs in Computing, Springer Nature, 2022.
1. K.-A. Mardal, M. E. Rognes and A. Tveito. Modeling excitable tissue: the EMI framework, Simula SpringerBriefs in Computing, Springer Nature, 2021.

Peer-reviewed chapters in books (selection)

11. H. Herlyng, M. Causemann, G. T. Einevoll, A. J. Ellingsrud, G. Halnes and M. E. Rognes, Modeling and simulation of electrodiffusion in dense reconstructions of cerebral tissue, in *50th Anniversary of the Woudschoten Conference*, Springer Verlag, 2026 (accepted)
10. M. E. Rognes, K.-A. Mardal, L. M. Valnes and V. Vinje, From brain physiology to brain physics, in *Mathematical modeling of the human brain II: from glymphatics to deep learning*, Springer Nature, 2026

9. M. Causemann, V. Vinje and M. E. Rognes, The pulsating brain: an interface-coupled fluid-poroelastic interaction model of the cranial cavity, in *Mathematical modeling of the human brain II: from glymphatics to deep learning*, Springer Nature, 2026
8. M. E. Rognes, An introduction to identifying velocity fields from contrast imaging via PDE-constrained optimization, in *Mathematical modeling of the human brain II: from glymphatics to deep learning*, Springer Nature, 2026
7. C. Daversin-Catty, J. P. Dean and M. E. Rognes, Finite element software and performance for network models with multipliers, Springer, 2024
6. M. Kuchta, K.-A. Mardal and M. E. Rognes. Solving the EMI equations using finite element methods. In Modeling excitable tissue: the EMI framework, edited by A. Tveito, K.-A. Mardal, and M. E. Rognes, Springer, 2021.
5. A. J. Ellingsrud, C. Daversin-Catty and M. E. Rognes. A cell-based model for ionic electrodiffusion in excitable tissue. In Modeling excitable tissue: the EMI framework, edited by A. Tveito, K.-A. Mardal, and M. E. Rognes, Springer, 2021.
4. G. Halnes, K. H. Pettersen, L. Øyehaug, M. E. Rognes and G. T. Einevoll. Astrocytic ion dynamics: implications for potassium buffering and liquid flow. In Computational Glioscience, Springer Series in Computational Neuroscience, 2019.
3. M. E. Rognes. Automated Testing of Saddle Point Stability Conditions. In Automated solution of differential equations by the finite element method, Springer-Verlag, 2012.
2. A. Logg, K. B. Ølgaard, M. E. Rognes and G. N. Wells. FFC: the FEniCS Form Compiler. In Automated solution of differential equations by the finite element method, Springer-Verlag, 2012.
1. L. Vynnytska, S. R. Clark and M. E. Rognes. Dynamic Simulations of Convection in the Earth's Mantle. In Automated solution of differential equations by the finite element method, Springer-Verlag, 2012.

Scientific presentations (selection)

I have given more than 100 scientific presentations (keynotes, plenary lectures, invited colloquia and seminars, and invited minisymposium talks) since 2007 including

2026* Invited talk, Computational Glioscience at the 28th meeting of the French Glial Cell Club, Seté, France
2026* 36th Mathematical Colloquium, Charles University, Prague, the Czech Republic
2026 Invited scientific lecture, FESTUM-PI, MAICh, Crete, Greece
2026 Plenary lecture, International Society for Magnetic Resonance in Medicine Workshop on fluids, flows, and clearance in the brain: what can we image and how should it be interpreted? Yale University, New Haven, USA
2026 MOX Colloquium, University of Milano, Milano, Italy
2025 Seminar at the Mittag-Leffler Institute, Sweden
2025 Minisymposium talk, ENUMATH 2025, Heidelberg, Germany
2025 Keynote lecture at the Center of Advanced Study opening conference: Mathematical challenges in brain mechanics, Center for Advanced Study, Norwegian Academy of Science and Letters
2025 Plenary lecture, 30th Biennial Numerical Analysis Conference, University of Strathclyde, Scotland
2025 Plenary lecture, XI International Conference on Coupled Problems in Science and Engineering (COUPLED 2025), Sardinia, Italy
2025 Computational and Applied Mathematics Seminar at Chalmers University of Technology, Sweden
2024 Plenary lecture: *Brain membranes and vasculature: a computational mathematics tale of dimensional gaps*, Nasjonalt matematikermøte (Norwegian Mathematics Meeting), Oslo, Norway
2024 Invited lecture: *Computational mathematics for the brain's waterscape*, ERC Synergy Grant NEMESIS Kick-off workshop, Institute Montpellier Alexandre Grothendieck, Montpellier, France
2024 European Society for Mathematical and Theoretical Biology Colloquium: *Brains in motion - computational modelling of the brain's waterscape*, online
2023 Minisymposium talk: *In-silico brain flow and transport*, the 2023 International Congress on Industrial and Applied Mathematics, Tokyo, Japan (online),
2023 Numerical Analysis and Scientific Computing Seminar: *Perivascular pathways and the dimension-2 gap*, Courant Institute, New York, US
2023 Applied Mathematics Seminar: *Perivascular pathways and the dimension-2 gap*, Brown University, Providence, US

2023 Oden Institute Seminar: *Brains in motion - computational modelling of the brain's waterscape*, University of Texas at Austin, TX, US
2023 CMOR Colloquium: *Perivascular pathways and the dimension-2 gap*, Rice University, Houston, TX, US
2023 Fluid Mechanics, Combustion, & Engineering Physics Seminar: *Brains in motion - computational modelling of the brain's waterscape*, UCSD, La Jolla, CA, USA
2022 Institute for Neural Computation Chalk Talk: *Mathematical modeling of human brain transport: from medical images to biophysical simulation*, San Diego Supercomputer Center, La Jolla, CA, USA
2022 Plenary lecture: *Such stuff as dreams are made on: a finite element tale of optimal transport and brain clearance*, 30 years of Acta Numerica, Banach Center at Bedlewo, Poland
2022 Semi-plenary lecture: *Mathematical modelling of the human brain*, 2022 ECCOMAS Congress, Oslo, Norway
2022 Pretty Porous Science Seminar, University of Stuttgart, Stuttgart, Germany
2022 LMS Seminar: *Mathematical modelling of human brain transport: from medical images to biophysical simulation*, Laboratoire de Mécanique des Solides (LMS), Ecole Polytechnique, Paris, France
2021 Invited lecture: *Modelling intracranial pressure, fluid flow and solute transport in surface perivascular space*, BrainH20 Symposium, Copenhagen, Denmark (online)
2021 Semi-plenary lecture: *Understanding the mechanisms of the brain's waterscape*, 2021 InterPore Conference (online)
2021 Invited talk: *Depressed brain cells - a numerical perspective*, 6th Oxford International Neuron and Brain Mechanics Workshop (online)
2021 Laboratoire Jacques-Louis Lions Seminar: *Numerical foundations of the brain's waterscape*, Sorbonne University, Paris, France (online)
2021 Plenary lecture: *Computational brainphatics*, 2021 SIAM Conference on Computational Science and Engineering (online)
2020 Invited lecture, Springer Partial Differential Equations and Applications webinar (online)
2020 Oxford Center for Industrial and Applied Mathematics Seminar, Mathematical Institute, Oxford, UK
2019 Invited talk: *UQ of parenchymal tracer distribution using random diffusion and convective velocity fields*, 5th biennial CSF Dynamics Symposium, Oslo, Norway
2019 Invited talk: *Nearly incompressible generalized poroelasticity - discretization and preconditioning*, International workshop in connection with Ragnar Winther's 70th birthday, Oslo, Norway
2018 Invited talk, 2nd Workshop on computational aspects of perfusion and flow in live tissue, Bergen
2017 Seminar: *The numerical waterscape of the brain*, Universite libre du Bruxelles, Belgium
2017 Invited talk, Biomechanics of living systems, from cells to organisms workshop, Oslo, Norway
2017 Plenary lecture, *Compatible discretizations in our hearts and minds*, European Conference on Numerical Mathematics and Advanced Applications (ENUMATH), Voss, Norway
2017 Invited talk: *Accurate numerical modelling of small collections of cardiac cells*, FEniCS'17, Luxembourg
2017 Plenary lecture: *Impact of high abstraction/high performance finite element software in biomedical computing*, 24th International Conference on Domain Decomposition Methods, Svalbard, Norway
2017 Keynote lecture: *The PDEs of our hearts and minds*, Women in PDEs, Karlsruhe, Germany
2016 Plenary lecture: *High-level abstractions, algorithms and applications in forward and inverse finite element solution of PDEs*, Nordic Seminar on Computational Mechanics, Chalmers, Sweden
2016 Invited Seminar, *Developing simulation technology to solve biomedical problems: analysis, implementation and applications*, University of Uppsala, Sweden
2016 Invited talk, *The FEniCS and Dolfin-adjoint projects*, Higher-order DG methods and finite element software for modern architectures workshop, University of Bath, UK

Outreach and popular science

2026 Public lecture, *Deltas of the Brain. Mathematics, Supercomputing and the Rhythms of Life* (YouTube), FESTUM-PI, Chania, Greece
2022 Invited article, *Waterscales: Mathematical and computational foundations for cerebral fluid flow*, European Mathematical Society Magazine
2018 Public lecture, *Hjernens vannveier*, Norwegian Academy of Technological Sciences, Bergen, Norway

2018 Invited talk for Young Scientist Contest Finalists: *The research life*, Research Council of Norway
2017 TEDx talk: *Mathematics that cures us* (on YouTube), TEDxOslo, Oslo, Norway
2017 Invited talk, *Making models of your brain's waterscape*, 2017 Cutting Edge Festival, Oslo
2016 Invited talk, *Time scales of mathematics: from basic research to societal application*, RCN workshop on the importance of mathematics for value creation, Lysaker, Norway